



DESN2000 SENG

Prelim Report

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Contents

1	Introduction and Report Scope	1
2	Problem Statement and Research Justification	1
2.1	Problem Statement	1
2.2	Research Justification	1
3	Solution Ideation	2
3.1	Research to Ideation Transition	2
3.2	Generated Technical Concepts	2
3.3	Ideation Outcome	3
3.4	Ideation Method	3
4	Concept Selection and Prioritisation	4
4.1	Selection Method	4
4.2	Decision Matrix	5
4.3	Selected Concept and Rationale	5
5	Product Backlog Structure	6
5.1	Epic Overview	6
5.2	User Stories	6
6	Acceptance Criteria	8
6.1	EP-01 Housemate Discovery and Trust	8
6.2	EP-02 Shared Household Living	10
6.3	EP-03 Accountability Extensions	15
6.4	EP-04 Property Management	16
7	Functional and Non-Functional Requirements	17
7.1	Functional Requirements	17
7.2	Non-Functional Requirements	17
8	User Story Dependencies	18
8.1	Dependency Table	18
8.2	Dependency Graph	20
9	First-Pass User Story Map	21
9.1	Sprint 1 Scope and Rationale	21

10 Preliminary Prototype	22
10.1 Prototype Scope	22
10.2 Prototype Flow	23
10.3 Prototype Screens	23
11 Technical Development Evolution	24
12 Innovation Compared to Existing Platforms	25
13 Prototype-to-User-Story Traceability	26
14 Prototype Test Plan	27
15 Conclusion	30
16 References	31
A Full User Story and Use Case Mapping	32
B Prototype Screens and Links	34
C Jira Evidence	35

1 Introduction and Report Scope

Previous report used surveys and interviews with four stakeholder groups to define the problem and develop personas. This report applies the Ideate stage to that evidence. It compares three technical concepts, selects one through a weighted decision matrix, and translates the selected direction into a Jira backlog and a preliminary prototype. The report focuses on the student journey from accommodation search to an agreed shared living arrangement. Property manager stories remain in the backlog as secondary scope.

2 Problem Statement and Research Justification

2.1 Problem Statement

As in the previous stage, the statement is:

Create a mobile platform which allows university students and shared accommodation residents to seamlessly manage household responsibilities, expenses and housemate communication, while also enabling property managers to streamline tenancy administration and tenant communication, to reduce conflict between tenants, streamline property management, and improve the overall shared living experience.

The statement covers student and property manager needs. This report narrows the preliminary prototype to the student path. A student searches for an option, checks verification, compares compatibility, starts contact in the app, and acknowledges a household rule. Property manager functions remain outside Sprint 1.

2.2 Research Justification

Table 1 links the strongest Part 1 evidence to a user need and a backlog response. This traceability keeps the proposed features grounded in the research.

Table 1. Research traceability.

Research evidence	User need	Backlog response
All 16 international respondents reported concern about rental scams. A further 43.75% had encountered misleading or false listing information.	The international student persona needs clear evidence about a profile or listing before making contact.	Show profile verification status in US-11 and clear listing details in US-13.
87.5% rated housemate compatibility as very or extremely important. A further 68.75% had experienced conflict with a previous housemate.	The long term housemate persona needs to identify practical mismatches before committing.	Compare living preferences in US-12.
44% identified communication difficulty as the hardest part of the search. More than 80% were willing to use preference based matching.	The international student persona needs first contact without early disclosure of personal details.	Keep first contact inside the app through US-09.
All surveyed domestic students relied on verbal household agreements and had experienced conflict over responsibilities. A further 66.67% wanted better chore tracking.	A new share house resident needs accepted expectations to remain visible after move in.	Store acknowledged household rules through US-02 and US-10.

3 Solution Ideation

3.1 Research to Ideation Transition

The evidence in Table 1 was reframed as four design needs. Table 2 sets the boundary for concept generation and later evaluation.

Table 2. Research issues and design needs.

Research issue	Design need
Scam concern and misleading information	Visible verification status and clear listing data
Conflict caused by poor living fit	Structured preference comparison
Risk during first contact	In app messaging before contact disclosure
Verbal household expectations	Recorded rules with acknowledgement

The resulting design question was as follows.

How might we help students move from an uncertain accommodation search to an informed shared living agreement?

Each concept was assessed against this question.

3.2 Generated Technical Concepts

The team generated three concepts with different technical boundaries. Table 3 gives the comparison. Tables 4 to 6 describe how each concept operates.

Table 3. Overview of the generated concepts.

Concept	Name	Main user task	Technical boundary
A	Verified Contact Gateway	Check status before first contact	Verification state and protected messaging
B	Explainable Compatibility Matcher	Compare practical living fit	Preference model and explainable score
C	Trust to Agreement Journey	Move from search to an acknowledged rule	Integrated flow across the core services

Table 4. Concept A. Verified Contact Gateway.

Aspect	Description
Purpose	Reduce uncertainty during first contact with an unfamiliar person.
Operation	The system records each verification check as <i>unverified</i> , <i>pending</i> , <i>verified</i> , <i>rejected</i> or <i>expired</i> . The interface shows the status and hides submitted evidence. Messaging does not expose personal contact details.
Research fit	Responds to scam concern and reluctance to contact unknown people.
Limitation	A verified identity does not establish compatibility or future conduct.

Table 5. Concept B. Explainable Compatibility Matcher.

Aspect	Description
Purpose	Help users identify practical mismatches before committing.
Operation	Users record living preferences and mark mandatory constraints. The matcher removes candidates who breach a mandatory constraint. It then calculates a weighted score and shows the factors that affected the result.
Research fit	Responds to the importance of daily living compatibility in the Part 1 findings.
Limitation	The score supports discussion but cannot predict a successful household.

Table 6. Concept C. Trust to Agreement Journey.

Aspect	Description
Purpose	Connect the accommodation decision with an explicit household agreement.
Operation	The flow links profile setup, status display, compatibility comparison, in app contact, and rule acknowledgement. Each step passes only the data needed by the next step. Later household tools remain outside the first release.
Research fit	Connects uncertainty before move in with unclear expectations after move in.
Limitation	Its wider scope raises delivery and privacy risk.

Concepts A and B isolate one decision in the user journey. Concept C integrates those decisions and adds agreement. All three remained candidates for the weighted evaluation.

3.3 Ideation Outcome

The ideation stage produced three candidates with different value and delivery risk. No concept was selected at this stage. All three entered the weighted decision matrix in Section 4.

3.4 Ideation Method

Required content

Name and explain the ideation method used by the team. Show how the method started from the problem definition and produced multiple ideas. Include evidence from the actual ideation activity.

Table 7. Ideation method template.

Ideation stage	Method used	Team activity	Output
<i>[Stage 1]</i>	<i>[Method]</i>	<i>[What the team did]</i>	<i>[Produced artefact]</i>
<i>[Stage 2]</i>	<i>[Method]</i>	<i>[What the team did]</i>	<i>[Produced artefact]</i>
<i>[Stage 3]</i>	<i>[Method]</i>	<i>[What the team did]</i>	<i>[Produced artefact]</i>

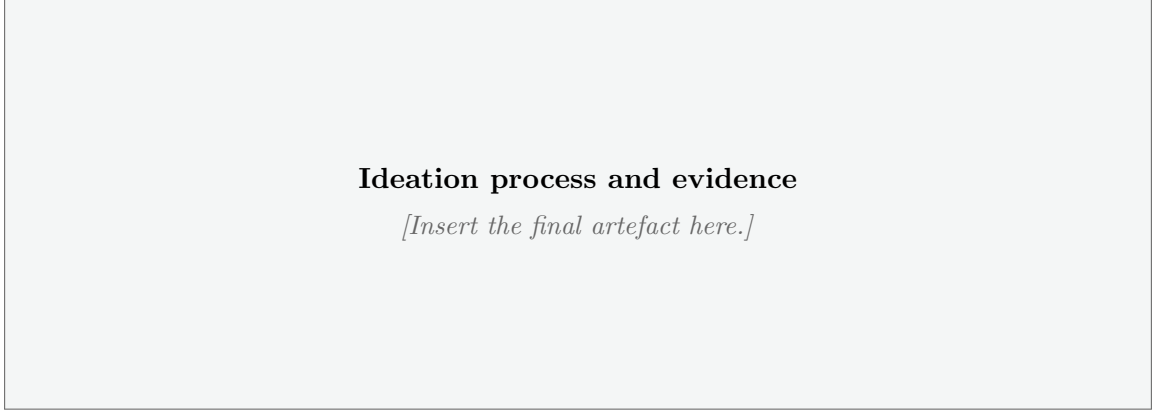


Figure 1. Ideation process and evidence.

4 Concept Selection and Prioritisation

4.1 Selection Method

The weighted decision matrix selected the concept. MoSCoW prioritisation then set the delivery scope. This separation prevented a broad concept from being treated as a commitment to deliver every related feature.

Each concept was rated from 1 to 5. A rating of 1 indicated poor fit. A rating of 3 indicated partial fit with material limits. A rating of 5 indicated full fit under the stated prototype assumptions. The weighted total was calculated as

$$S_j = \sum_i w_i r_{ij}$$

where w_i is the criterion weight and r_{ij} is the rating for concept j .

Table 8. Selection criteria and weights.

Criterion	Weight	Rationale
Research evidence fit	30%	Measures direct traceability to the Part 1 findings and personas.
User risk reduction	20%	Measures how much evidenced user risk the concept reduces.
Feasibility within ten weeks	20%	Measures whether the team can prototype and revise the concept within the schedule.
Prototype testability	15%	Measures whether users can complete observable tasks in a clickable flow.
Innovation beyond isolated tools	10%	Measures whether the concept creates a research grounded journey rather than copying one existing function.
Accessibility and privacy risk	5%	Measures whether foreseeable risks can be controlled. An unacceptable risk remains a rejection condition.

The final ratings in Table 9 must match the scoring evidence in Figure 2.

Week 6 scoring record reconstructed for team confirmation			
Rating order: research fit; risk reduction; feasibility; testability; innovation; accessibility/privacy.			
Team member	Concept A	Concept B	Concept C
Nuodi Liu	4, 4, 5, 4, 3, 4	4, 4, 4, 5, 4, 4	5, 5, 3, 4, 5, 3
Qian Cao	4, 4, 5, 4, 3, 4	4, 4, 4, 4, 4, 4	5, 5, 3, 4, 5, 3
Chien-Kai Wang	4, 4, 5, 4, 3, 4	4, 4, 4, 5, 4, 4	5, 5, 2, 4, 5, 3
Pranav Shanker	4, 4, 5, 4, 3, 4	4, 4, 4, 5, 4, 4	5, 5, 3, 5, 5, 3
Final consensus	4, 4, 5, 4, 3, 4	4, 4, 4, 5, 4, 4	5, 5, 3, 4, 5, 3
Consensus discussion. Concept A received the highest feasibility score because it has the narrowest technical scope. Concept B received a testability score of 5 because users can complete and assess the compatibility task independently. Concept C received scores of 3 for feasibility and privacy because its integrated flow adds dependencies and data-sharing decisions. The team selected Concept C for its research fit and risk reduction, and used MoSCoW prioritisation to limit the first release.			

Figure 2. Week 6 concept-selection evidence: independent ratings and consensus discussion.

4.2 Decision Matrix

The weighted total is the sum of each rating multiplied by its weight. The maximum score is 5.00.

Table 9. Weighted decision matrix.

Selection criterion	Weight	Concept A	Concept B	Concept C
Research evidence fit	30%	4	4	5
User risk reduction	20%	4	4	5
Feasibility within ten weeks	20%	5	4	3
Prototype testability	15%	4	5	4
Innovation beyond isolated tools	10%	3	4	5
Accessibility and privacy risk	5%	4	4	3
Weighted total	100%	4.10	4.15	4.35

Concept C ranks first with 4.35. It fits the research and covers more of the target journey. Its lower feasibility and privacy rating exposes the main delivery risk.

4.3 Selected Concept and Rationale

Concept C was selected because it links the strongest research findings in one user path. Concept A stops after safer first contact. Concept B stops after the match decision. Concept C continues to an acknowledged household rule.

The preliminary prototype uses a narrow release slice to control scope. Table 10 aligns the priority of every story with the Sprint 1 path shown later in the report.

Table 10. MoSCoW priorities for preliminary prototype delivery.

Priority	User stories	Delivery decision
Must have in Sprint 1	US-02, US-09, US-10, US-11, US-12, US-13	These stories form one testable path from search to an acknowledged rule.
Should have later	US-01, US-03, US-04, US-05, US-06, US-14	These stories extend the agreement into routine household follow through.
Could have	US-07, US-08, US-15	These stories add supporting communication or reputation value without blocking the core path.
Exploratory only	US-16, US-17, US-18	Clickable property manager screens communicate future role coverage, but they are excluded from the evaluated Sprint 1 path and test plan.

Only the Must have row defines Sprint 1. The same release slices are used in the backlog tables and must match Jira. Exploratory screens do not change a story’s delivery priority.

5 Product Backlog Structure

5.1 Epic Overview

The backlog is organised into four Epics that follow the shape of the selected Trust-to-Agreement Co-living Journey. Two Epics deliver the MVP journey before and after move-in, while the remaining two hold extension work together with the secondary property management scope so that the first sprint stays focused. Every story carries a stable identifier, which means a later change of sprint or priority does not break its link to the matching use case in Appendix A.

Table 11. Epic structure of the product backlog.

Epic ID	Epic name	Epic goal and value	Linked user stories
EP-01	Housemate Discovery and Trust	Help accommodation seekers judge whether a listing or a potential housemate can be trusted before any commitment is made.	US-09, US-11, US-12, US-13
EP-02	Shared Household Living	Make household expectations explicit and keep everyday responsibilities visible after move-in.	US-01 to US-08 and US-10
EP-03	Accountability Extensions	Strengthen follow-through on accepted responsibilities once the core journey is validated.	US-14, US-15
EP-04	Property Management	Give property managers a direct channel to publish listings and communicate with their tenants.	US-16, US-17, US-18

5.2 User Stories

Each story uses the standard user, goal, and value structure. The scope column follows Table 10.

Table 12. EP-01 Housemate Discovery and Trust user stories.

Story ID	User story	Scope	Jira ticket
US-09	As an international student, I want to message a potential housemate in the app before sharing contact details, so that I can make first contact with less exposure to scams or harassment.	Sprint 1	ST4-35
US-11	As an international student, I want to see a profile's verification status, so that I can assess the identity signal before contacting that person.	Sprint 1	ST4-38
US-12	As an accommodation seeker with an irregular schedule, I want to compare potential housemates against my living preferences, so that I can identify practical mismatches before committing.	Sprint 1	ST4-39
US-13	As a student moving out of home, I want to filter listings by location and cost, so that I can narrow the search to suitable options.	Sprint 1	ST4-43

Table 13. EP-02 Shared Household Living user stories.

Story ID	User story	Scope	Jira ticket
US-01	As a new share house resident, I want to assign rotating chores and view completion history, so that each contribution remains visible.	Later sprint	ST4-23
US-02	As an international student in a new household, I want to view the accepted house rules in one place, so that I can understand what the household expects.	Sprint 1	ST4-24
US-03	As a new share house resident, I want housemates to review a proposed chore system before it becomes active, so that the roster reflects a shared decision.	Later sprint	ST4-26
US-04	As a share house resident, I want to record a shared expense and who paid it, so that the cost can be divided among the relevant housemates.	Later sprint	ST4-28
US-05	As a share house resident, I want to view outstanding balances, so that I know who owes money and how much.	Later sprint	ST4-29
US-06	As an international student, I want to divide a utility bill among selected housemates, so that my share is clear.	Later sprint	ST4-32
US-07	As a resident who is often away, I want to post household notices in a dedicated space, so that important updates are not lost in chat.	Backlog	ST4-33
US-08	As a resident, I want to raise a household issue without showing my identity to other residents, so that I can report a sensitive concern with less social pressure.	Backlog	ST4-34
US-10	As a resident with different noise habits from my housemates, I want to propose a house rule and record each acknowledgement, so that the active rule is clear during a dispute.	Sprint 1	ST4-36

Table 14. EP-03 Accountability Extensions user stories.

Story ID	User story	Scope	Jira ticket
US-14	As a resident, I want an in app reminder when my chore or payment is overdue, so that I can act before a housemate must follow up.	Later sprint	ST4-25
US-15	As an accommodation seeker, I want to read reviews from confirmed former housemates, so that I can consider past shared living experience before choosing a match.	Backlog	ST4-41

Table 15. EP-04 Property Management user stories.

Story ID	User story	Scope	Jira ticket
US-16	As a property manager, I want to record rent status and send overdue reminders, so that I can follow up late payments efficiently.	Secondary	ST4-30
US-17	As a property manager, I want to publish an accurate rental listing, so that students can assess the property and enquire through the platform.	Secondary	ST4-40
US-18	As a property manager, I want to send an update to every current tenant at a property, so that important information reaches the correct audience.	Secondary	ST4-37

6 Acceptance Criteria

Every user story in the backlog has measurable acceptance criteria written in the Given When Then form, so each table in this section can be executed directly as a test skeleton. Each story covers the successful path together with at least one error scenario and one edge scenario. The same criteria are recorded on the matching Jira ticket, whose key appears in each caption so that the backlog in Jira and this report stay verifiable against each other. The tables are grouped under the Epics defined in Table 11.

6.1 EP-01 Housemate Discovery and Trust

Table 16. Acceptance criteria for US-09 (Jira ST4-35).

Acceptance ID	Scenario	Given	When	Then
AC-US09-01	Success	two users are connected on the platform	they open a conversation	they can send and receive messages through a secure in-app system without either party's personal contact details being visible
AC-US09-02	Success	both users mutually agree to share contact details	they both opt in	the system exchanges their contact details between them
AC-US09-03	Success	a user feels unsafe	they block or report another user from the messaging interface	the reported user is flagged for review and the conversation is ended
AC-US09-04	Error	a user attempts to include personal contact details in a message	the system detects this	it warns the user and recommends using the opt-in contact sharing feature instead
AC-US09-05	Error	a user attempts to contact someone who has blocked them	they try to send a message	the system prevents the message from being sent without revealing that they have been blocked
AC-US09-06	Edge case	a user's account is flagged as unverified	they attempt to initiate a conversation	the system displays a warning to the recipient that the sender is unverified before they choose to respond

Table 17. Acceptance criteria for US-11 (Jira ST4-38).

Acceptance ID	Scenario	Given	When	Then
AC-US11-01	Success	a user has submitted a valid form of identification	the verification is approved	their profile displays a verified identity badge visible to all other users
AC-US11-02	Success	a user is searching for housemates	they apply the verified only filter	only profiles with a verified identity badge appear in search results
AC-US11-03	Success	a user views an unverified profile	they open it	a visible warning indicator is displayed clearly at the top of the profile
AC-US11-04	Error	a user submits an expired or unreadable identification document	the system processes it	the verification is rejected and the user is notified with instructions to resubmit a valid document
AC-US11-05	Error	a user attempts to access the platform's core matching features without completing verification	they try to proceed	the system prompts them to complete verification first
AC-US11-06	Edge case	a verified user's identification document expires	the system detects the expiry	it notifies the user to resubmit updated documentation to maintain their verified status

Table 18. Acceptance criteria for US-12 (Jira ST4-39).

Acceptance ID	Scenario	Given	When	Then
AC-US12-01	Success	a user has completed their lifestyle preference profile	they search for housemates	results are automatically ranked by a compatibility score calculated from matching preferences
AC-US12-02	Success	a user wants to evaluate a potential housemate	they open their profile	a detailed compatibility summary is displayed showing areas of strong match and potential differences
AC-US12-03	Success	a potential housemate views the user's profile	they open it	they receive a notification that someone has viewed their profile
AC-US12-04	Error	a user has not completed their lifestyle preference profile	they attempt to search for compatible housemates	the system prompts them to complete their profile before search results can be personalised
AC-US12-05	Error	a user sets contradictory preferences (e.g. early riser and late sleeper simultaneously)	they attempt to save	the system flags the conflict and prompts them to review their selections
AC-US12-06	Edge case	no housemates match a user's strict preference filters	the search returns zero results	the system suggests broadening one or more filters and displays the closest available matches

Table 19. Acceptance criteria for US-13 (Jira ST4-43).

Acceptance ID	Scenario	Given	When	Then
AC-US13-01	Success	a student enters their university name as a reference point	they search for listings	results are displayed ranked by proximity to the selected campus
AC-US13-02	Success	a student applies filters for maximum weekly rent, distance from campus, number of bedrooms and move-in date	they search	only listings matching all applied filters are displayed
AC-US13-03	Success	a student finds a listing of interest	they open it	they can view affordability indicators including weekly rent in AUD, estimated cost per person, a comparison to the local suburb average and a Budget/Moderate/Premium affordability rating
AC-US13-04	Error	a student searches for listings in an area with no available results	the search returns empty	the system notifies them and suggests broadening their search radius or adjusting filters
AC-US13-05	Error	a listing contains inaccurate photos or descriptions	a student reports it	the listing is flagged for review and temporarily hidden from search results if multiple reports are received
AC-US13-06	Edge case	a student's preferred university is not listed in the system	they search	the system allows them to enter a custom address or suburb as an alternative reference point

6.2 EP-02 Shared Household Living

Table 20. Acceptance criteria for US-01 (Jira ST4-23).

Acceptance ID	Scenario	Given	When	Then
AC-US01-01	Success	a user has created a household and added housemates	they set up a chore roster with task assignments and a rotation schedule	the system distributes chores evenly and notifies each housemate of their assigned tasks
AC-US01-02	Success	a housemate has been assigned a chore	they mark it as complete	the system logs the completion with a timestamp and updates the participation summary
AC-US01-03	Success	a user wants to review contributions	they access the chore history	they can see a full log of who completed what and when
AC-US01-04	Error	a user attempts to create a roster without assigning all housemates	they submit	the system prompts them to assign at least one task to each housemate before activating
AC-US01-05	Error	a chore deadline has passed without being marked complete	the system checks for overdue tasks	it flags the chore as overdue and notifies the assigned housemate
AC-US01-06	Edge case	a housemate leaves the household	their assigned chores become unassigned	the system alerts the house admin to reassign those tasks

Table 21. Acceptance criteria for US-02 (Jira ST4-24).

Acceptance ID	Scenario	Given	When	Then
AC-US02-01	Success	a housemate has added household rules and expectations	an international student opens the household rules section	they can view all rules in a single clearly organised list
AC-US02-02	Success	a user has set their language preference to a non-English language	they view household rules	the rules are displayed in their preferred language or simplified English
AC-US02-03	Success	a user wants clarification on a specific rule	they tap on it	they can view an expanded description or explanation provided by the housemate who created it
AC-US02-04	Error	a user attempts to save a rule without a title or description	they submit	the system prompts them to complete all required fields before saving
AC-US02-05	Error	no household rules have been added yet	a user opens the rules section	the system displays a prompt encouraging housemates to add expectations
AC-US02-06	Edge case	a rule is written in English but the user's preferred language is different	the translation service is unavailable	the system displays the rule in simplified English as a fallback

Table 22. Acceptance criteria for US-03 (Jira ST4-26).

Acceptance ID	Scenario	Given	When	Then
AC-US03-01	Success	a user wants to establish a chore system	they submit a proposed chore system including task assignments and schedules	all housemates receive a notification to review the proposal
AC-US03-02	Success	all housemates have reviewed the proposal	they all approve it	the chore system becomes active and assignments are distributed accordingly
AC-US03-03	Success	a housemate wants to suggest changes	they submit feedback on the proposal	the proposer receives a notification and can revise and resubmit
AC-US03-04	Error	a housemate does not respond to the proposal within 7 days	the deadline passes	the system sends a reminder and flags the proposal as awaiting response
AC-US03-05	Error	a user submits a proposal without assigning tasks to all housemates	they attempt to submit	the system prompts them to assign at least one task per housemate
AC-US03-06	Edge case	one housemate rejects the proposal	the rejection is submitted	the system notifies the proposer and pauses activation until a revised proposal is approved by all

Table 23. Acceptance criteria for US-04 (Jira ST4-28).

Acceptance ID	Scenario	Given	When	Then
AC-US04-01	Success	a user has paid for a shared expense	they log it with a description, amount, category and payer	the system automatically calculates each selected housemate's share and notifies them
AC-US04-02	Success	an expense has been logged	a housemate views their dashboard	they can see all expenses they are involved in and their individual amount owed
AC-US04-03	Success	a housemate has paid their share	the payer marks it as settled	the balance between those two housemates updates automatically
AC-US04-04	Error	a user submits an expense without entering an amount	they attempt to save	the system prompts them to complete all required fields
AC-US04-05	Error	a user attempts to split an expense among housemates not in their household	they submit	the system only allows splitting among confirmed household members
AC-US04-06	Edge case	a housemate leaves the household with an outstanding balance	their account is removed	the system retains their balance record and flags it as unresolved for the remaining housemates

Table 24. Acceptance criteria for US-05 (Jira ST4-29).

Acceptance ID	Scenario	Given	When	Then
AC-US05-01	Success	expenses have been logged between housemates	a user opens their balance dashboard	they can see a clear summary of who owes them money and how much
AC-US05-02	Success	a housemate has paid their share	the user marks the balance as settled	the dashboard updates automatically to reflect the new balance
AC-US05-03	Success	balances exist across multiple housemates	the user views the dashboard	all balances are displayed individually per housemate
AC-US05-04	Error	a user attempts to mark a balance as settled that has already been settled	they submit	the system notifies them that the balance is already cleared
AC-US05-05	Error	no expenses have been logged yet	a user opens the balance dashboard	the system displays a prompt to add the first shared expense
AC-US05-06	Edge case	a balance is disputed by a housemate	they flag it	the system marks the balance as disputed and notifies both parties to resolve it before it can be settled

Table 25. Acceptance criteria for US-06 (Jira ST4-32).

Acceptance ID	Scenario	Given	When	Then
AC-US06-01	Success	a utility bill has been received	a user enters the bill type, total amount and billing period	the system calculates and assigns each housemate's share based on the selected split method
AC-US06-02	Success	a bill has been logged	all housemates view the shared bill history	they can see all submitted bills and their individual amounts owed
AC-US06-03	Success	a housemate believes a bill is incorrect	they flag it as disputed	all housemates are notified and the bill is marked as under review
AC-US06-04	Error	a user submits a bill without entering a total amount	they attempt to save	the system prompts them to complete all required fields
AC-US06-05	Error	a user attempts to split a bill among zero housemates	they submit	the system requires at least two participants before allowing submission
AC-US06-06	Edge case	a utility bill covers a period during which a housemate was absent	the bill is split	the system allows the user to manually exclude or adjust that housemate's share for that billing period

Table 26. Acceptance criteria for US-07 (Jira ST4-33).

Acceptance ID	Scenario	Given	When	Then
AC-US07-01	Success	a user wants to post a notice	they create a notice with a title, description and optional image	it is immediately visible to all housemates on the noticeboard
AC-US07-02	Success	a notice has been posted	a housemate wants to respond	they can leave a limited reaction or reply directly on the notice
AC-US07-03	Success	a notice has an expiry date set	the date passes	the notice is automatically archived and removed from the active noticeboard
AC-US07-04	Error	a user attempts to post a notice without a title	they submit	the system prompts them to add a title before posting
AC-US07-05	Error	a user sets an expiry date in the past	they submit	the system rejects the date and prompts them to select a future date
AC-US07-06	Edge case	a notice receives no interaction for 30 days and has no expiry date set	the system checks for inactive notices	it prompts the creator to either renew or archive the notice

Table 27. Acceptance criteria for US-08 (Jira ST4-34).

Acceptance ID	Scenario	Given	When	Then
AC-US08-01	Success	a user wants to raise an issue	they submit it with the anonymous option selected	the issue is posted to the shared household space without revealing the submitter's name or profile
AC-US08-02	Success	an issue has been posted	housemates view it	they can vote on whether they agree it is a concern
AC-US08-03	Success	all parties have agreed the issue is resolved	the submitter or a majority vote marks it as resolved	the issue is closed and archived
AC-US08-04	Error	a user attempts to submit an issue without a description	they submit	the system prompts them to add a description before posting
AC-US08-05	Error	a user attempts to identify an anonymous submitter through the system	they try to access submitter information	the system does not reveal any identifying information under any circumstance
AC-US08-06	Edge case	an anonymous issue receives no votes or responses within 14 days	the system checks for inactive issues	it sends a reminder to all housemates to review and respond

Table 28. Acceptance criteria for US-10 (Jira ST4-36).

Acceptance ID	Scenario	Given	When	Then
AC-US10-01	Success	a user wants to propose a house rule	they submit it with a title, description and category	all housemates receive a notification to review and acknowledge it
AC-US10-02	Success	all housemates have acknowledged a proposed rule	the last acknowledgement is submitted	the rule is officially added to the household rules list with a timestamp
AC-US10-03	Success	a user wants to reference a rule during a dispute	they link it to a household issue	the rule is displayed alongside the issue for all housemates to see
AC-US10-04	Error	a user attempts to submit a rule without a title or category	they submit	the system prompts them to complete all required fields
AC-US10-05	Error	a housemate attempts to edit or remove a rule without majority approval	they submit the change	the system rejects it and notifies all housemates that a change has been requested pending approval
AC-US10-06	Edge case	a rule has not been acknowledged by one housemate after 7 days	the system checks for pending acknowledgements	it sends a reminder to that housemate to review and acknowledge the rule

6.3 EP-03 Accountability Extensions

Table 29. Acceptance criteria for US-14 (Jira ST4-25).

Acceptance ID	Scenario	Given	When	Then
AC-US14-01	Success	a chore deadline has passed and the chore has not been marked complete	the system checks for overdue tasks	it automatically sends an in-app notification to the assigned housemate
AC-US14-02	Success	the house admin wants to manually remind a housemate	they send a nudge	the assigned housemate receives an in-app notification without any personal contact details being exchanged
AC-US14-03	Success	a housemate marks an overdue chore as complete after receiving a reminder	they submit the completion	the system updates the chore log and removes the overdue flag
AC-US14-04	Error	a housemate has disabled notifications	a chore becomes overdue	the system still flags the chore as overdue within the app and displays it prominently on their dashboard
AC-US14-05	Edge case	a housemate is temporarily removed from the roster (e.g. on leave)	their chore deadline passes	the system does not send them an overdue reminder and reassigns the task to the next available housemate

Table 30. Acceptance criteria for US-15 (Jira ST4-41).

Acceptance ID	Scenario	Given	When	Then
AC-US15-01	Success	a confirmed tenancy period has ended	a user submits a written review and star rating for their previous housemate	the review is displayed on that housemate's profile in chronological order
AC-US15-02	Success	a user views a housemate's profile	they scroll to the reviews section	they can see all reviews with the overall star rating displayed prominently at the top
AC-US15-03	Success	a housemate receives a review	it is posted	they receive a notification and can submit a single written response
AC-US15-04	Error	a user attempts to submit a review before a confirmed tenancy period has ended	they try to submit	the system prevents submission and notifies them that reviews can only be left after a confirmed tenancy
AC-US15-05	Error	a user attempts to submit a second review for the same housemate from the same tenancy	they submit	the system rejects the duplicate and notifies the user
AC-US15-06	Edge case	a review is flagged as inappropriate by multiple users	the system detects multiple reports	the review is temporarily hidden pending platform review and the reviewer is notified

6.4 EP-04 Property Management

Table 31. Acceptance criteria for US-16 (Jira ST4-30).

Acceptance ID	Scenario	Given	When	Then
AC-US16-01	Success	a property manager has set a rent amount and due date for a tenant	the due date approaches	the system automatically sends a rent reminder to that tenant
AC-US16-02	Success	a tenant's payment has been recorded	the property manager views the dashboard	the tenant's status updates to "paid"
AC-US16-03	Success	the due date has passed without payment being recorded	the system checks payment status	the tenant's status updates to "overdue" and the property manager is notified
AC-US16-04	Error	a property manager attempts to set a rent due date without entering a rent amount	they submit	the system prompts them to complete all required fields
AC-US16-05	Error	a tenant's contact details are incomplete	the system attempts to send a reminder	the system flags the tenant's profile as incomplete and alerts the property manager
AC-US16-06	Edge case	a tenant disputes a rent amount	they flag it through the app	the system marks the payment status as disputed and notifies the property manager to review

Table 32. Acceptance criteria for US-17 (Jira ST4-40).

Acceptance ID	Scenario	Given	When	Then
AC-US17-01	Success	a property manager has completed a listing with photos, description, rent price, available date and house rules	they publish it	it immediately appears in student search results based on location and filters
AC-US17-02	Success	a student finds a listing they are interested in	they send an enquiry	the property manager receives a notification and the message is stored in their centralised inbox
AC-US17-03	Success	a listing's available date has passed	the system checks active listings	the listing is automatically removed from search results and the property manager is notified
AC-US17-04	Error	a property manager attempts to publish a listing without photos or a rent price	they submit	the system prompts them to complete all required fields before publishing
AC-US17-05	Error	a student attempts to enquire about a listing that has already been filled	they submit the enquiry	the system notifies them that the listing is no longer available
AC-US17-06	Edge case	a listing has been live for 60 days without being filled	the system checks listing age	it prompts the property manager to review and update the listing details

Table 33. Acceptance criteria for US-18 (Jira ST4-37).

Acceptance ID	Scenario	Given	When	Then
AC-US18-01	Success	a property manager wants to send an announcement	they compose it with a title, description, category and optional attachment	it is delivered to all tenants in the selected property
AC-US18-02	Success	an announcement has been sent	a tenant opens the app	they receive a push notification and can view the announcement in their notification centre
AC-US18-03	Success	a property manager wants to schedule a recurring announcement	they set a recurrence schedule	the system automatically sends the announcement at the defined intervals
AC-US18-04	Error	a property manager attempts to send an announcement without selecting a property	they submit	the system prompts them to select at least one property before sending
AC-US18-05	Error	a scheduled announcement fails to send	the system detects the failure	it retries delivery and notifies the property manager if the issue persists
AC-US18-06	Edge case	a tenant has not viewed a high priority announcement within 24 hours	the system checks read receipts	it sends a follow-up push notification to that tenant

7 Functional and Non-Functional Requirements

7.1 Functional Requirements

Required content

Define what the system must do. Link every requirement to one or more user stories and state how the requirement will be verified.

Table 34. Functional requirements template.

Requirement ID	Functional requirement	Related stories	Verification method
FR-01	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>
FR-02	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>
FR-03	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>

7.2 Non-Functional Requirements

Required content

Include measurable requirements for privacy, security, accessibility, performance and usability. Add other categories where they are relevant to the solution.

Table 35. Non-functional requirements template.

Requirement ID	Category	Measurable requirement	Verification method
NFR-01	Privacy	<i>[Privacy target]</i>	<i>[Verification method]</i>
NFR-02	Security	<i>[Security target]</i>	<i>[Verification method]</i>
NFR-03	Accessibility	<i>[Accessibility target]</i>	<i>[Verification method]</i>
NFR-04	Performance	<i>[Performance target]</i>	<i>[Verification method]</i>
NFR-05	Usability	<i>[Usability target]</i>	<i>[Verification method]</i>

8 User Story Dependencies

Each user story maps one to one onto a goal-level use case, so the dependency analysis is presented with use case identifiers and applies directly to the corresponding stories. The analysis also covers four supporting use cases together with two seeded data sources, because they enable several stories at once rather than belonging to a single story. The full mapping between stories and use cases is recorded in Appendix A.

8.1 Dependency Table

Table 36. Dependency types used in the analysis.

Type	Meaning
Required	The dependent use case cannot proceed until the prerequisite is satisfied.
Required data	The dependent use case needs data that the prerequisite creates or maintains.
Required data (OR)	Any one of the listed prerequisites supplies the needed data.
Conditional	The dependency applies only in a specific situation, such as showing a verified badge.
Optional	The relationship adds support but is not required for the main success path.

Table 37. Dependencies within EP-02 Shared Household Living.

Dependent use case	Depends on	Type	Reason
UC-01 Chore roster	SUP-02 Household membership	Required	Only members of an active household can view or manage its chore roster.
UC-02 View household rules	SUP-02 Household membership	Required	Rules belong to a specific household and only members can view or manage them.
UC-02 View household rules	UC-10 Agreed house rules	Required data	Rules must be set and agreed before they can be displayed in a member's preferred language.
UC-03 Propose chore system	UC-01 Chore roster	Required data	A structured chore system cannot be proposed or activated without an existing roster to build on.
UC-03 Propose chore system	SUP-02 Household membership	Required	All approving housemates must be identified members of the same household.
UC-04 Log and split expenses	SUP-02 Household membership	Required	Expenses and each member's share must belong to an authorised household.
UC-05 Track balances	UC-04 Shared expenses	Required data	Balances are computed from logged expenses, so at least one expense must exist to show who owes what.
UC-06 Split utility bills	UC-04 Shared expenses	Required	A utility bill is itself a shared expense, so UC-06 reuses the logging and cost-splitting capability of UC-04 rather than any pre-existing expense data.
UC-07 Household notices	SUP-02 Household membership	Required	The noticeboard is scoped to a single household and only members can post or read notices.
UC-08 Anonymous issues	SUP-02 Household membership	Required	The issue must belong to the reporter's household while hiding their identity from other residents.
UC-08 Anonymous issues	UC-10 Agreed house rules	Optional	Existing house rules give issues a shared reference point during disputes, but an issue can be raised without them.

Table 38. Dependencies within EP-01 Housemate Discovery and Trust.

Dependent use case	Depends on	Type	Reason
UC-09 Secure messaging	UC-11 Verified profiles	Optional	Core secure messaging works on its own with personal contact details hidden, while UC-11 adds the flag that marks unverified counterparties for extra caution.
UC-11 Verified profiles	SUP-04 Verification status or DATA-02 Seeded profiles	Required data (OR)	A stored verification status or a seeded sample profile must exist before a badge can be shown.
UC-12 Lifestyle filtering	UC-11 Verified profiles	Required	Compatibility matching is only offered between verified profiles to preserve trust, and a match does not grant household access.
UC-13 Search listings	UC-17 Managed listings or DATA-01 Seeded listings	Required data (OR)	At least one published listing, whether managed or seeded, must exist before a student can search it.

Table 39. Dependencies within EP-03 Accountability Extensions and EP-04 Property Management.

Dependent use case	Depends on	Type	Reason
UC-14 Overdue reminders	UC-01 Chore roster or UC-04 Shared expenses	Required data (OR)	A reminder needs an overdue chore assignment or an overdue expense to reference.
UC-14 Overdue reminders	SUP-03 Push notifications	Required	Automatic overdue reminders are delivered through the notification service.
UC-15 Housemate reviews	SUP-02 Household or tenancy membership	Required	A review may only be submitted after a confirmed tenancy or shared living arrangement has ended.
UC-16 Rent tracking	SUP-02 Tenancy membership	Required	Rent status needs a confirmed relationship between tenant and property, and selecting an applicant is not enough.
UC-16 Rent tracking	SUP-03 Push notifications	Optional	Push notifications deliver due date and overdue rent reminders outside the active app.
UC-17 Manage listings	SUP-04 Verification status	Conditional	Verification is required only before a verified badge is shown, and an unverified or pending listing may still be saved.
UC-18 Tenant announcements	SUP-02 Tenancy membership	Required	The system needs a confirmed current tenant audience for the selected property.
UC-18 Tenant announcements	SUP-03 Push notifications	Optional	Push notifications deliver the announcement outside the active app.

8.2 Dependency Graph

Figure 3 shows the validated dependency graph for the complete backlog. Solid arrows mark required relationships while dotted arrows mark optional or conditional support. The diamond nodes are OR points where any single source is sufficient, which allows the Sprint 1 prototype to rely on seeded data instead of the secondary property management workflow.

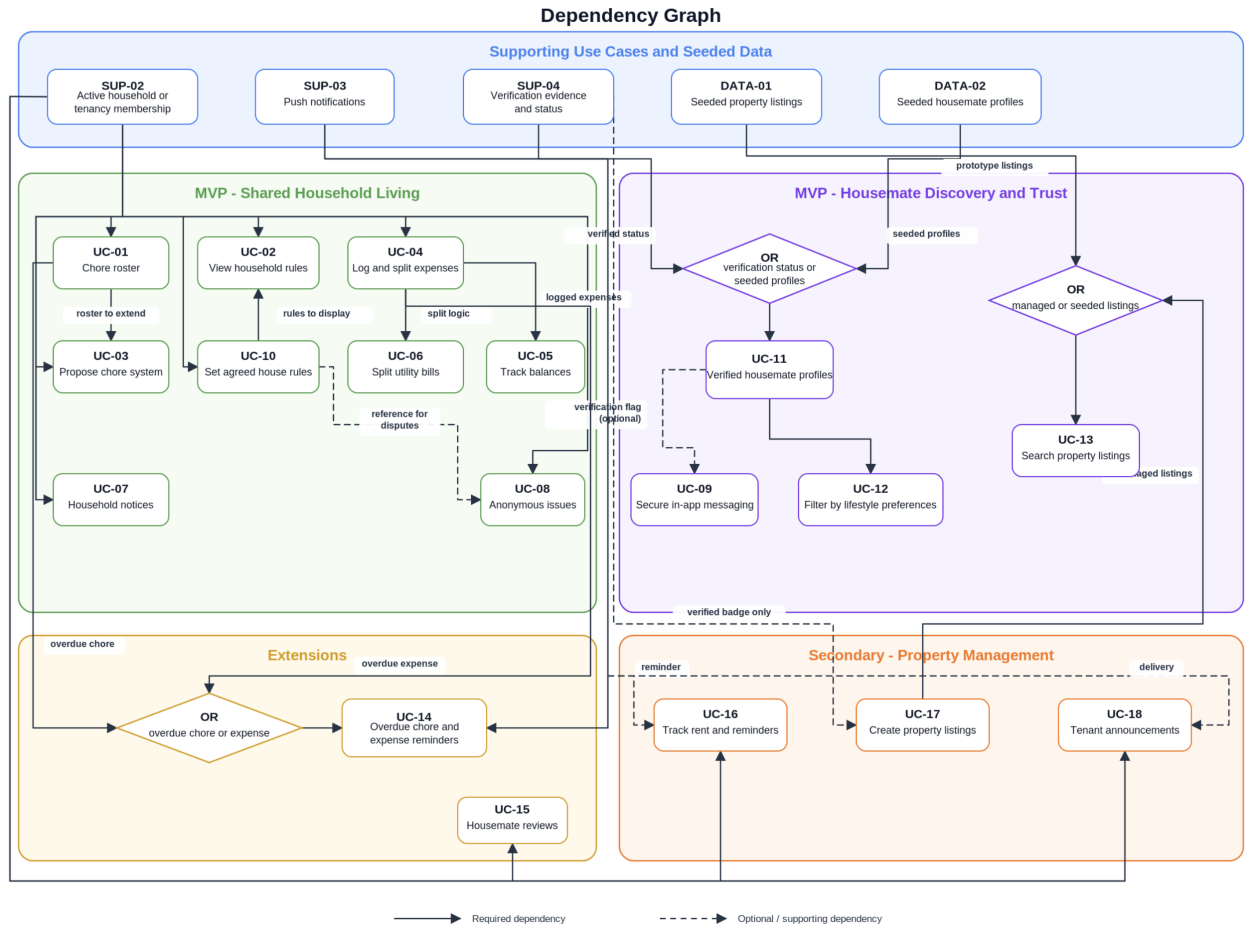


Figure 3. Use case dependency graph for the product backlog.

9 First-Pass User Story Map

Table 40 summarises the release slice used to define Sprint 1. The complete team map, including personas, all four activities, backbone tasks and release priorities, is provided in Figure 9 in Appendix A.

Table 40. First-pass user story map for the Trust-to-Agreement journey.

Release slice	Find a place	Evaluate people	Contact safely	Set expectations	Manage the shared home
User task	Filter by campus and budget	Read verification and lifestyle fit	Verify identity and message in-app	Propose, understand and agree rules	Coordinate chores, bills and issues
Sprint 1	US-13	US-11, US-12	US-09	US-10, US-02	–
Later sprint	–	US-15	–	–	US-01, US-03–US-08, US-14
Secondary backlog	US-17	–	–	–	US-16, US-18

9.1 Sprint 1 Scope and Rationale

Sprint 1 contains US-13, US-11, US-12, US-09, US-10 and US-02. Together they form a coherent end-to-end slice rather than six disconnected features: a student narrows affordable properties, evaluates verified and compatible housemates, completes the trust gate required for private messaging, and then accesses explicit household expectations. This sequence tests the selected concept’s central claim that trust information should continue from discovery into the first shared-living agreement.

The slice is narrow enough to test with seeded listings and profiles, while still covering search, decision support, safety, communication and rule comprehension. Chores, expenses, reminders and property management remain visible as later backlog directions, but they are not required to complete the Sprint 1 evaluation path.

10 Preliminary Prototype

10.1 Prototype Scope

The preliminary prototype is a functional high-fidelity HTML, CSS and JavaScript proof of concept for students and young professionals who are searching for shared accommodation, particularly users entering an unfamiliar household or communicating in a second language. It uses realistic seeded properties, housemate profiles, messages and house rules so that participants can complete tasks without entering personal data.

Table 41. Prototype scope and fidelity decisions.

Decision	Prototype implementation
Core task path	Search and filter properties (US-13), assess verification and lifestyle compatibility (US-11 and US-12), authenticate before messaging (US-09), and understand household rules (US-02 and US-10).
Fidelity	Functional high fidelity. Navigation, filters, validation, authentication gates, messaging and multilingual rule interactions are clickable rather than represented by static wireframes.
Visual rationale	Warm photography, warm white surfaces, peach highlights and sage trust signals follow the mood board. Compact layouts and plain labels support repeated use rather than a marketing-style experience.
Prototype data	Seeded listings and profiles provide repeatable test conditions. Changes last for the current page session and reset on refresh, so every participant begins from the same state.
Test environment	The prototype is hosted as a web application and can also run locally. Desktop is the primary assessment viewport, with responsive navigation and layouts for smaller screens.

This fidelity is appropriate because the rubric requires observable, testable interactions linked to user stories. A clickable interface allows the team to measure completion, errors and comprehension, while avoiding the cost and privacy risk of a production backend at this stage. The prototype does not perform real identity checks, production encryption, database persistence or external translation services. Property filtering, compatibility calculation and unanimous rule approval run locally against seeded data, while the phone check uses a demonstration code. The prototype therefore tests the interaction logic and user comprehension without claiming production security or independently authenticated housemate accounts.

10.2 Prototype Flow

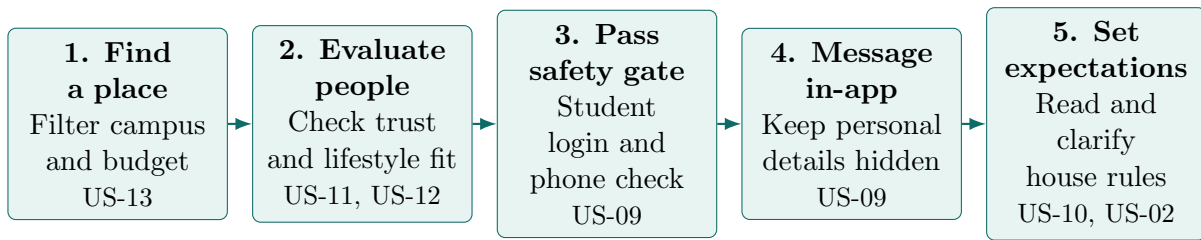


Figure 4. Sprint 1 prototype flow from accommodation search to shared expectations.

10.3 Prototype Screens

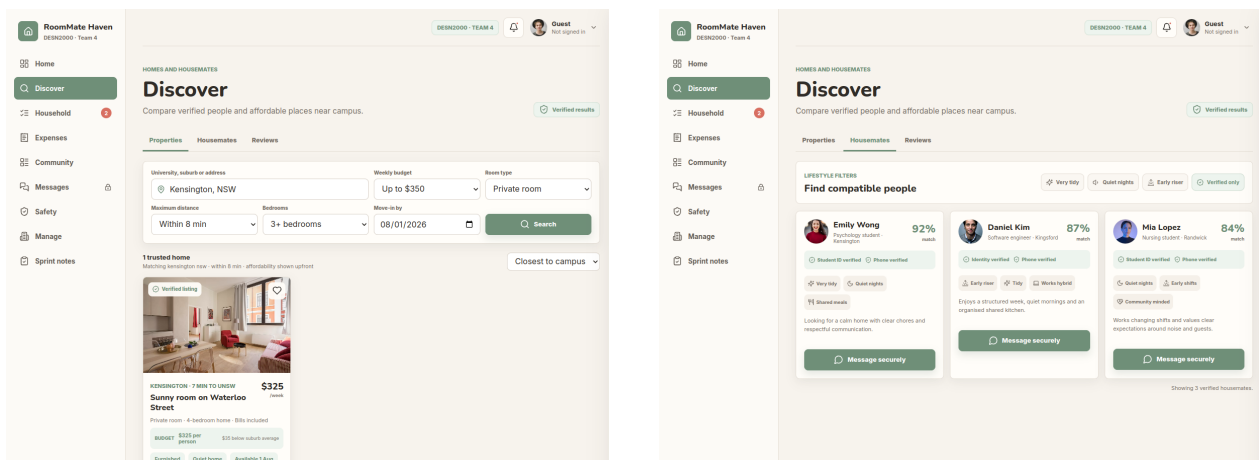


Figure 5. Discovery screens for property affordability and profile verification.

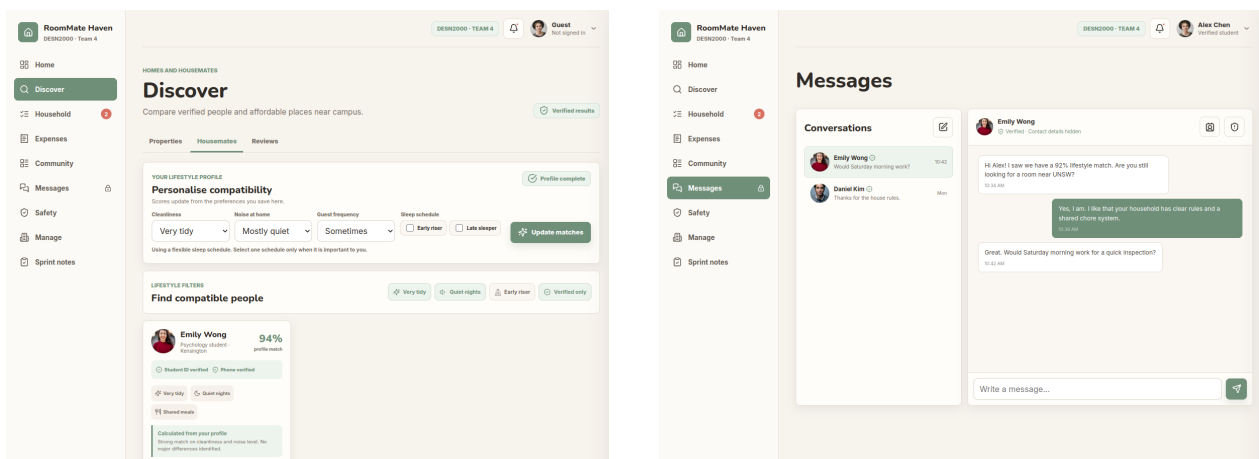
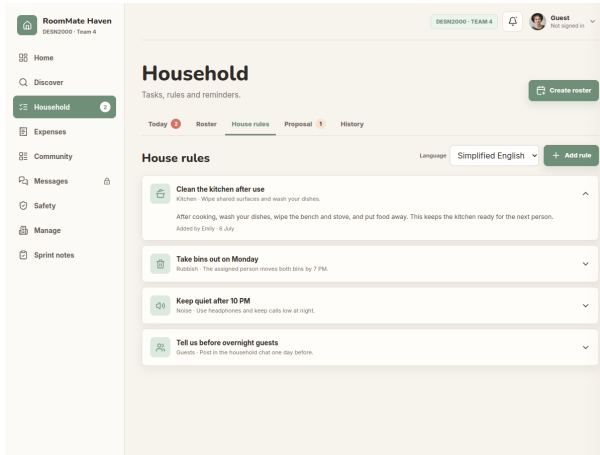
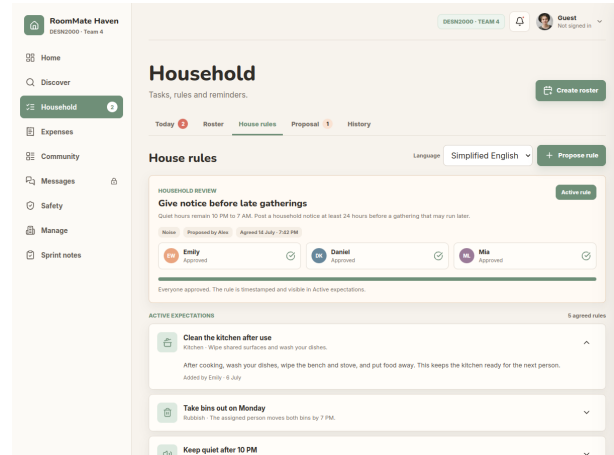


Figure 6. Trust and communication screens for comparing and contacting housemates.



S-05: Household rules. All expectations appear in one location with language controls and expandable explanations.



S-06: Unanimous rule approval. A proposal becomes active only after Emily, Daniel and Mia have all approved it.

Figure 7. Shared-expectation screens for accessible and explicit household rules.

The screens demonstrate both the normal path and observable system feedback: multi-criteria property filters update and sort the result set, lifestyle preferences recalculate compatibility scores, contradictory schedules are rejected, messaging remains locked until verification, personal contact details trigger a warning, language changes update rule content, and a proposed rule remains inactive until every housemate approves it.

11 Technical Development Evolution

During Weeks 3 to 7, the proposed solution gradually evolved as new research findings and team discussions shaped the design. Initially, the project focused on creating a general student accommodation platform where users could browse available properties and contact landlords. However, the user research indicated that students were more concerned about trust and housemate compatibility than simply finding available rooms.

Based on these findings, the team shifted the design towards a compatibility-focused platform. Lifestyle preference profiles and compatibility comparison became the core feature, supported by verified listings and guided in-app messaging to improve trust during communication. Finally, a lightweight co-living agreement was added to encourage discussion of household expectations before students moved in.

This iterative process ensured that the final concept directly addressed the problems identified during the Empathise and Define stages while remaining achievable within the project scope. The resulting Trust-to-Agreement journey described in Section 4 reflects this progression from a broad accommodation search tool to a student-centred decision path.

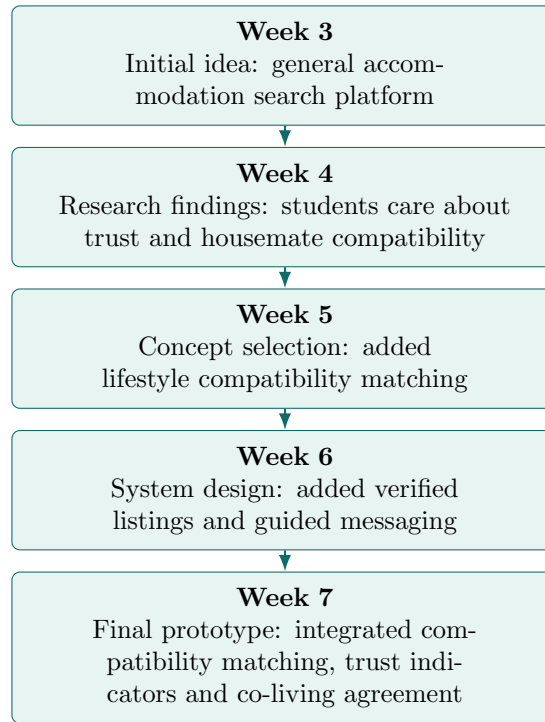


Figure 8. Technical development evolution from Weeks 3 to 7.

12 Innovation Compared to Existing Platforms

Existing accommodation platforms such as Airbnb, Flatmates.com.au and Domain mainly focus on helping users search for available properties and contact landlords or housemates. While these platforms provide useful listing information, they offer limited support for evaluating lifestyle compatibility or preparing users for shared living before moving in.

Our proposed platform goes beyond accommodation search by integrating features specifically designed for university students. It combines lifestyle compatibility matching, verified trust indicators, guided in-app messaging and a simple pre-move-in co-living agreement. These features help users make more informed decisions and reduce potential conflicts before committing to a shared living arrangement.

Rather than replacing existing accommodation platforms, the proposed solution complements them by addressing the trust, compatibility and shared-living challenges identified through our user research. Table 42 summarises how the selected Sprint 1 features compare with three widely used services reviewed during the competitive analysis.

Table 42. Feature comparison between existing accommodation platforms and our proposed platform.

Feature	Airbnb	Flatmates.com.au	Domain	Our Platform
Property listings	✓	✓	✓	✓
Verified listings	Partial	Partial	✓	✓
Lifestyle compatibility matching	–	Limited	–	✓
Guided in-app messaging	Limited	Basic	Basic	✓
Pre-move-in co-living agreement	–	–	–	✓
Student-focused experience	–	Partial	–	✓

Airbnb supports short-term stays and host verification but does not target long-term student housemate matching or pre-move-in household agreements [1]. Flatmates.com.au lists shared accommodation and basic profile filters, yet its compatibility support remains limited and it does not provide a structured agreement workflow before move-in [2]. Domain offers verified rental listings for the Australian market, but its search flow stops at property discovery and does not extend into housemate fit or shared-living preparation [3]. Our platform is the only option in this comparison that combines all six capabilities in one student-focused path.

13 Prototype-to-User-Story Traceability

Each screen is mapped to the user story and acceptance criteria that it demonstrates. Table 43 identifies the exact interaction evidence and keeps production-system boundaries separate from prototype task completion. The same screen identifiers are used by the test plan in Section 14.

Table 43. Prototype-to-user-story traceability.

Screen ID	Prototype screen	Story ID	Acceptance ID	Testable interaction	Link
S-01	Multi-criteria property search	US-13	AC-US13-01-04, AC-US13-06	Combine university or suburb, budget, room type, distance, bedrooms and move-in date; results update, sort by campus distance, show affordability and provide a broadening suggestion when no exact match exists.	Open
S-02	Verified housemate results	US-11	AC-US11-01, AC-US11-02	Identify identity and phone badges, then inspect the verified-only filter used by the result set.	Open
S-03	Profile-derived compatibility ranking	US-12	AC-US12-01, AC-US12-02, AC-US12-04-06	Edit the lifestyle profile to recalculate and re-rank scores, inspect the explanation, submit incomplete or conflicting preferences for validation, and broaden strict filters after a zero-result state.	Open
S-04	Authentication and secure messages	US-09	AC-US09-01, AC-US09-04	Attempt to message while signed out, complete login and the phone check, send a normal message, then enter an email or phone number to trigger blocking and a privacy warning.	Open
S-05	Multilingual household rule list	US-02	AC-US02-01, AC-US02-02, AC-US02-03	View every rule in one list, change the language, and expand a rule to reveal the housemate's explanation.	Open
S-06	Rule proposal and unanimous approval	US-10, US-02	AC-US10-01, AC-US10-02, AC-US10-04, AC-US02-01	Submit an incomplete rule for validation, send a complete proposal to every housemate, approve or request changes as each reviewer, and confirm the rule enters the active list with an agreement timestamp only after all approvals.	Open

14 Prototype Test Plan

The prototype will be evaluated through five moderated usability tests that follow the Sprint 1 path. Each test uses five participants and a think-aloud protocol. The facilitator records completion, time, errors and interpretation rather than helping participants through the interface. Because the prototype uses only seeded data and a demonstration verification code, the sessions do not require participants to disclose identification, phone numbers or real household information.

Table 44. Test overview and coverage of Sprint 1 stories.

Test	Goal	Stories	Acceptance criteria	Responsible
T1	Narrow properties with combined search criteria	US-13	AC-US13-01-04, AC-US13-06	Pranav Shanker
T2	Recognise verified housemate profiles	US-11	AC-US11-01, AC-US11-02	Nicky
T3	Build a profile and use recalculated compatibility	US-12	AC-US12-01, AC-US12-02, AC-US12-04-06	Liam
T4	Complete authentication and message safely	US-09	AC-US09-01, AC-US09-04	Qian Cao
T5	Propose, review and activate a house rule	US-02, US-10	AC-US02-01-03, AC-US10-01, AC-US10-02, AC-US10-04	Liam

Table 45. Test T1. Narrow properties with combined search criteria.

Aspect	Description
Test goal	Verify that a student can combine location, affordability and availability criteria to identify a suitable property.
Related stories	US-13; S-01
Acceptance criteria	AC-US13-01, AC-US13-02, AC-US13-03, AC-US13-04 and AC-US13-06
Target user	Student moving near UNSW with a fixed weekly budget.
Task scenario	“Find a private room in Kensington within eight minutes of UNSW, in a household with at least three bedrooms, available by 1 August, for no more than \$350 per week. Explain its affordability rating.”
Success criteria	$\geq 4/5$ users enter all six criteria, reach the single matching listing and explain its Budget indicator in ≤ 3 minutes. A deliberately impossible combination must produce the no-result explanation and broadening action.
Expected errors	Users overlook a second-row filter, interpret the move-in date in the wrong direction, or confuse household bedrooms with available rooms.
Data to collect	Completion rate, time on task, filters changed, selected listing, affordability interpretation and recovery from the empty state.
Responsible	Pranav Shanker

Table 46. Test T2. Recognise verified housemate profiles.

Aspect	Description
Test goal	Verify that visible identity signals help users recognise which profiles have been checked.
Related stories	US-11; S-02
Acceptance criteria	AC-US11-01, AC-US11-02
Target user	International student assessing unfamiliar potential housemates.
Task scenario	“Find housemates whose identity and phone have been verified. Show the control you would use to avoid unverified results and explain what evidence you relied on.”
Success criteria	$\geq 4/5$ users locate the verified-only control and correctly identify both badge types in ≤ 90 seconds. Participants must explain that the badge is a trust signal, not a guarantee of compatibility or safety.
Expected errors	Users miss the badge, confuse identity and phone verification, or assume a high match score means identity verified.
Data to collect	Badge identification accuracy, explanation quality, time on task and points of visual confusion.
Responsible	Nicky

Table 47. Test T3. Build a profile and use recalculated compatibility.

Aspect	Description
Test goal	Verify that users can create a complete preference profile and understand how it changes compatibility ranking.
Related stories	US-12; S-03
Acceptance criteria	AC-US12-01, AC-US12-02, AC-US12-04, AC-US12-05 and AC-US12-06
Target user	Student who values a tidy home and quiet nights.
Task scenario	“Save a very tidy, mostly quiet profile with occasional guests and a flexible schedule. Identify the best match and explain the score. Then select both Early riser and Late sleeper, and finally apply all three strict lifestyle filters.”
Success criteria	$\geq 4/5$ users save the complete profile, identify Emily as the 94% top match and cite a profile-derived reason in ≤ 3 minutes. The contradictory schedule must be rejected, and the zero-result state must offer the closest-match recovery action.
Expected errors	Users treat the score as a safety rating, do not notice that ranking changes after saving, or overlook the schedule conflict message.
Data to collect	Profile completion, ranking interpretation, conflict recovery, empty-state recovery, time on task and reasons given.
Responsible	Liam

Table 48. Test T4. Complete authentication and message safely.

Aspect	Description
Test goal	Verify that messaging is inaccessible before verification and that personal contact details are intercepted.
Related stories	US-09; S-04
Acceptance criteria	AC-US09-01, AC-US09-04
Target user	International student making first contact with a potential housemate.
Task scenario	“Starting signed out, open Messages, sign in with the demo account, verify the phone using code 2468 and send a question about an inspection. Then try to send alex@example.com.”
Success criteria	$\geq 4/5$ users complete login and phone verification and send the normal message in ≤ 3 minutes. Messaging must stay locked beforehand, and the email attempt must not appear in the thread while the privacy warning is visible.
Expected errors	Users close the login modal and expect guest messaging, cannot find the phone-check screen, or misread the warning as a successfully sent message.
Data to collect	Completion at each gate, time on task, failed code entries, message outcome and warning comprehension.
Responsible	Qian Cao

Table 49. Test T5. Propose, review and activate a house rule.

Aspect	Description
Test goal	Verify that a household rule remains a proposal until every housemate approves it, while existing expectations remain accessible.
Related stories	US-02, US-10; S-05 and S-06
Acceptance criteria	AC-US02-01, AC-US02-02, AC-US02-03, AC-US10-01, AC-US10-02 and AC-US10-04
Target user	Student living in a shared apartment for the first time and using English as a second language.
Task scenario	“Open the quiet-hours explanation, then propose a new late-gathering rule. First submit it without a category, then complete it. Record one change request, revise it, and approve as Emily, Daniel and Mia.”
Success criteria	$\geq 4/5$ users trigger the required-field message, send the complete proposal, understand that a change request pauses activation, and reach the Active rule state only after the third approval in ≤ 4 minutes. The rule must then appear in Active expectations with all names and a timestamp.
Expected errors	Users confuse a proposal with an active rule, expect one approval to be sufficient, or miss the revise-and-resubmit action after a requested change.
Data to collect	Validation errors noticed, proposal-state comprehension, approval order, time on task and confirmation that the active timestamp was found.
Responsible	Liam

After each round the team will compile results into one summary, rank observed issues by severity and feed them into the next prototype iteration. Failed acceptance criteria will be recorded by identifier. The next technical iteration would replace seeded local data with persistent services, run rule approval through independently authenticated housemate accounts, and connect verification to a secure identity provider.

15 Conclusion

Section purpose

Summarise the selected concept, the Sprint 1 scope, the preliminary prototype and the next validation step. Do not introduce new evidence in the conclusion.

[Insert the conclusion after the report content is complete.]

16 References

References

- [1] Airbnb. (n.d.). *Help Centre: How Airbnb works*. <https://www.airbnb.com.au/help/article/362>
- [2] Flatmates.com.au. (n.d.). *Find a flatmate or room*. <https://flatmates.com.au/>
- [3] Domain. (n.d.). *Rental property search*. <https://www.domain.com.au/rent/>

A Full User Story and Use Case Mapping

Figure 9 presents the complete User Story Map used to organise personas, product activities, backbone tasks and release priorities. It provides the visual planning context for the detailed traceability that follows, where each stable story identifier maps to one goal-level use case in Table 50.



Figure 9. Complete User Story Map showing personas, activities, backbone tasks and release priorities.

Table 50. User story to use case mapping.

Story	Epic	Use case	Primary actor	Successful outcome	Scope	Website screenshot
US-01	EP-02	UC-01 Set up and manage chore roster	Domestic student / shared-house resident	The user can create, assign and rotate chores among housemates and track completion history.	MVP	View
US-02	EP-02	UC-02 View and manage household rules	International student / shared-house resident	The user can view all household rules and expectations in their preferred language in a single location.	MVP	View
US-03	EP-02	UC-03 Propose and approve chore system	Domestic student	All housemates have reviewed and approved a structured chore system before it becomes active.	MVP	View
US-04	EP-02	UC-04 Log and split shared expenses	Shared-house resident	The user can log a shared expense and the system automatically calculates and assigns each housemate's share.	MVP	View
US-05	EP-02	UC-05 Track outstanding balances	International student / shared-house resident	The user can view a real-time summary of who owes what and mark balances as settled.	MVP	View
US-06	EP-02	UC-06 Split utility bills	International student / shared-house resident	The user can input a utility bill and split it equally or by usage among selected housemates.	MVP	View
US-07	EP-02	UC-07 Post and view household notices	Shared-house resident	The user can post a notice visible to all housemates on a dedicated noticeboard separate from general messaging.	MVP	View
US-08	EP-02	UC-08 Raise anonymous household issues	Domestic student / shared-house resident	A household issue is posted to the shared space without revealing the submitter's identity if the anonymous option is selected.	MVP	View
US-09	EP-01	UC-09 Send secure in-app messages	International student / shared-house resident	The user can send and receive messages without either party's personal contact details being visible.	MVP	View
US-10	EP-02	UC-10 Set and view agreed house rules	Shared-house resident	All housemates have acknowledged a proposed house rule before it becomes active and can reference it during disputes.	MVP	View
US-11	EP-01	UC-11 View verified housemate profiles	International student / accommodation seeker	The user can view verified profiles with identity badges and filter search results to show only verified profiles.	MVP	View
US-12	EP-01	UC-12 Filter housemates by lifestyle preferences	Shared-house resident / accommodation seeker	The user can set lifestyle preference filters and view a compatibility score and summary for each potential housemate.	MVP	View
US-13	EP-01	UC-13 Search and filter property listings	Domestic student / accommodation seeker	The user can search for listings by university proximity and apply filters for rent, distance, bedrooms and move-in date.	MVP	View
US-14	EP-03	UC-14 Receive overdue chore and expense reminders	Shared-house resident	The user receives automatic in-app reminders when a chore or expense payment is overdue.	Extension	View
US-15	EP-03	UC-15 Read and submit housemate reviews	Shared-house resident / accommodation seeker	The user can read reviews left by previous housemates and submit a review after a confirmed tenancy ends.	Extension	View
US-16	EP-04	UC-16 Track rent and send reminders	Property manager	Rent status is recorded per tenant and overdue tenants receive automatic reminders.	Secondary	View
US-17	EP-04	UC-17 Create and manage property listings	Property manager	A complete property listing is published and visible to students searching for accommodation.	Secondary	View
US-18	EP-04	UC-18 Send tenant announcements	Property manager	An announcement is delivered to all tenants of a selected property with read receipt tracking.	Secondary	View

Supporting use cases enable several user goals at once and therefore do not map to a single story. Two seeded data sources let the prototype demonstrate the discovery features without implementing the secondary property workflow first.

Table 51. Supporting use cases.

ID	Supporting use case	Primary actor	Successful outcome
SUP-01	Manage account and role access	Any registered user	The user is authenticated and receives the correct student, resident, or property-manager permissions.
SUP-02	Establish active household or tenancy membership	Resident / property manager / selected tenant	Membership is activated only after a household invitation or tenancy is confirmed.
SUP-03	Deliver push notifications	System	Chore reminders, expense reminders, announcement deliveries, and issue notifications can reach users outside the active app.
SUP-04	Manage verification evidence and status	User / property manager / system	Evidence is stored and the related profile receives a pending, verified, or rejected status.

Table 52. Supporting prototype data.

Data ID	Data source	Purpose
DATA-01	Seeded property listings	Provides published sample listings when the MVP prototype does not implement the secondary property-listing workflow.
DATA-02	Seeded housemate profiles	Provides sample verified and unverified profiles for compatibility matching and verification testing in the MVP prototype.

B Prototype Screens and Links

Table 53. Prototype access information.

Item	Access details
Hosted prototype	RoomMate Haven on AWS Elastic Beanstalk
Demo account	Student email: <code>z5555555@ad.unsw.edu.au</code> ; password: <code>studentdemo</code>
Phone verification	Select “Send code” and enter the demonstration code 2468 .
Reset behaviour	Refresh the page to restore all seeded data and return the prototype to a repeatable test state.
Screenshot evidence	GitHub screenshot folder containing US-01 to US-18 evidence

Table 54. Direct evidence states for Sprint 1 user stories.

Story	Screenshot file	Hosted evidence state
US-02	US-02_household-rules.png	Open capture state
US-09	US-09_secure-in-app-messaging.png	Open capture state
US-10	US-10_agreed-noise-rules.png	Open capture state
US-11	US-11_verified-housemate-profiles.png	Open capture state
US-12	US-12_lifestyle-housemate-filters.png	Open capture state
US-13	US-13_property-search-filters.png	Open capture state

C Jira Evidence

Appendix content

Insert Jira backlog exports, screenshots or links showing Epics, user stories, acceptance criteria and sprint assignment.